

Solving Quadratic Equations: Choose a Method, Read the Roots

Answer Key

Algebra 1

Quick Answers

- | | |
|--------------------------------------|------------------------------------|
| 1. 7, -7 | 7. $\frac{5}{2}$, -1 |
| 2. 2, 3 | 8. 7, -12 |
| 3. $-2 + \sqrt{3}$, $-2 - \sqrt{3}$ | 9. 1, -5 |
| 4. 7, -1 | 10. $\frac{1}{3}$, $-\frac{3}{2}$ |
| 5. 0, 3 | 11. 2.62 |
| 6. 0, 4 | 12. 9, 21 |
-

1. $x^2 = 49$ — **square roots.** The variable appears only as x^2 , with no linear term, so take square roots directly; factoring $x^2 - 49 = (x - 7)(x + 7)$ works too but is slower.

$$x = \pm\sqrt{49}$$

$x = 7 \text{ or } x = -7$

Both signs are required: dropping the negative root is the standard slip here.

2. $x^2 - 5x + 6 = 0$ — **factoring.** Monic with small integers: look for two numbers multiplying to 6 and adding to -5 , namely -2 and -3 .

$$(x - 2)(x - 3) = 0$$

Zero-product property: $x - 2 = 0$ or $x - 3 = 0$.

$x = 2 \text{ or } x = 3$

3. $x^2 + 4x + 1 = 0$ — **quadratic formula.** No integer pair multiplies to 1 and adds to 4, so factoring fails; with $a = 1$, $b = 4$, $c = 1$:

$$x = \frac{-4 \pm \sqrt{16 - 4}}{2} = \frac{-4 \pm \sqrt{12}}{2} = \frac{-4 \pm 2\sqrt{3}}{2} = -2 \pm \sqrt{3}$$

$x = -2 + \sqrt{3} \text{ or } x = -2 - \sqrt{3}$

The discriminant 12 is positive but not a perfect square — that is exactly the signature of a quadratic that will not factor over the integers.

4. $2(x - 3)^2 = 32$ — **square roots**. The whole variable part sits inside one square, so isolate it and take roots; expanding first would create a trinomial you would then have to factor back.

$$\begin{aligned}(x - 3)^2 &= 16 \\ x - 3 &= \pm 4 \\ x &= 3 + 4 = 7 \quad \text{or} \quad x = 3 - 4 = -1\end{aligned}$$

$$\boxed{x = 7 \text{ or } x = -1}$$

5. Ball from ground level, $h = -16t^2 + 48t$. Both terms share $-16t$, so factoring is fastest:

$$-16t^2 + 48t = -16t(t - 3) = 0$$

$$\boxed{t = 0 \text{ or } t = 3}$$

Now read the roots. $t = 0$ is the instant of the throw — the ball is at ground level because it has not left yet. $t = 3$ is the landing. The ball is therefore in the air for $\boxed{3 \text{ seconds}}$. Neither root is thrown away here: one is the start of the story, the other is the answer to the question asked.

6. $3x^2 = 12x$ — **factoring after moving everything to one side**.

$$3x^2 - 12x = 0 \implies 3x(x - 4) = 0$$

$$\boxed{x = 0 \text{ or } x = 4}$$

Common wrong answer: if they wrote only $x = 4$, they divided both sides by x . Dividing by a variable that can be zero deletes the root $x = 0$ — move everything to one side and factor instead.

7. $2x^2 - 3x - 5 = 0$ — **quadratic formula (or factoring by grouping)**. With $a = 2$, $b = -3$, $c = -5$: $\Delta = 9 + 40 = 49$, a perfect square.

$$x = \frac{3 \pm 7}{4}$$

$$\boxed{x = \frac{5}{2} \text{ or } x = -1}$$

Because the discriminant is a perfect square, this one does factor: $(2x - 5)(x + 1) = 0$ gives the same roots. Either method is efficient; the formula needs no searching.

8. Garden: $w^2 + 5w - 84 = 0$. Find factors of -84 that add to 5: 12 and -7 .

$$(w + 12)(w - 7) = 0$$

$$\boxed{w = 7 \text{ or } w = -12}$$

Interpret: w is a width, and a width cannot be negative, so $w = -12$ is rejected as extraneous *in this context* even though it solves the equation. The garden is $\boxed{7 \text{ m}}$ wide (and $7 + 5 = 12$ m long, giving $7 \cdot 12 = 84$ square metres ✓).

9. $3(x+2)^2 = 27$ — **square roots**.

$$(x+2)^2 = 9$$

$$x+2 = \pm 3$$

$$x = -2 + 3 = 1 \quad \text{or} \quad x = -2 - 3 = -5$$

$$x = 1 \text{ or } x = -5$$

10. $6x^2 + 7x - 3 = 0$ — **factoring with $a \neq 1$ (grouping)**. $a \cdot c = 6(-3) = -18$; find factors of -18 adding to 7 : 9 and -2 .

$$6x^2 + 9x - 2x - 3 = 0$$

$$3x(2x+3) - 1(2x+3) = 0$$

$$(3x-1)(2x+3) = 0$$

$$x = \frac{1}{3} \text{ or } x = -\frac{3}{2}$$

The quadratic formula gives the same pair; grouping is quicker once $a \cdot c$ splits cleanly.

11. **Ball from 5 feet up**, $h = -16t^2 + 40t + 5$. It does not factor, so use the formula with $a = -16$, $b = 40$, $c = 5$:

$$\Delta = 40^2 - 4(-16)(5) = 1600 + 320 = 1920$$

$$t = \frac{-40 \pm \sqrt{1920}}{-32} = \frac{40 \mp \sqrt{1920}}{32}$$

The two roots are $t = \frac{40 + \sqrt{1920}}{32} \approx 2.6193$ and $t \approx -0.1193$. Time started when the ball was released, so a negative time is before the throw and cannot be the landing: reject it. The ball lands after

$$2.62 \text{ seconds}$$

(nearest hundredth). **Common wrong answer:** $t \approx -0.12$ comes from taking the other branch of the $\pm$ — check every root against the situation before writing it down.

12. **Synthesis — break-even for $P = -n^2 + 30n - 189$** . (a) Multiply through by -1 (or factor out -1) so the leading coefficient is positive, then factor: $n^2 - 30n + 189 = 0$ needs factors of 189 adding to 30 , namely 9 and 21 .

$$-(n-9)(n-21) = 0$$

$$n = 9 \text{ or } n = 21$$

(b) The manager is wrong. Break-even happens at exactly 9 and 21 sweatshirts, and 20 is between them. Because the parabola opens downward ($a = -1 < 0$), P is *positive* strictly between the roots: at $n = 20$, $P = -400 + 600 - 189 = 11$ dollars, a small profit rather than break-even. Outside the roots (fewer than 9 or more than 21) the club loses money. *Note for the grader: part (b) is a written interpretation, judged by you; only the two break-even roots are machine-verified.*